Supplementary Material

Spatio-temporal wall-actuation modes and cycle-mean viscous dissipation in a lid-driven cavity

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Overview

This Supplementary Material accompanies the manuscript *"Spatio-temporal wall-actuation modes and cycle-mean viscous dissipation in a lid-driven cavity"*. It contains detailed numerical formulations, complete network architecture specifications, cell- and seed-level campaign datasets for all sensitivity campaigns, grid-convergence calculations, the wall-shear estimator analysis, and reproducibility guidelines. The material is organized so that the main text can be read as a fluid-mechanics paper: every result that concerns the identification procedure, the variational landscape, the sensitivity campaigns or the numerical verification is documented here, while the main text retains only the physical formulation, the actuation modes, the dissipation comparison and its interpretation. All section numbers in this document are prefixed with "S" to distinguish them from main-text sections, and all figures are numbered S1, S2, ... independently of the main text. The tables are numbered in the same way, so that Table S1 of this document is distinct from Table 1 of the main text. Cross-references to the main manuscript use the notation (Main Sec., Main Fig. X or Main Table X).

1 S1. Detailed PINN Implementation

1.1 S1.1 Network Architecture Specifications

The physics-informed optimization employs two separate fully connected neural networks operating in double-precision (float64) arithmetic throughout: a *flow network* that represents the velocity and pressure fields $(u, v, p)(x, y, t; \widetilde{Re})$, and a *control network* that represents the spatio-temporal Fourier coefficients $a_{mn}(\widetilde{Re})$. The two networks share the same input normalization but have different widths and depths, reflecting the different complexity of the flow field and the low-dimensional control space.

Input normalization. The spatial coordinates $(x, y) \in [0, L_x] \times [0, 1]$ and time $t \in [0, 1]$ are fed directly to the networks. The Reynolds number is normalized to

$$\widetilde{Re} = \frac{Re - 550}{450} \in [-1, 1], \tag{1}$$

so that the training range $Re \in [100, 1000]$ maps to $[-1, 1]$, improving numerical conditioning of the gradient computations.

Flow network. Six hidden layers, each with 64 neurons, using hyperbolic tangent (tanh) activation functions. Inputs: $(x, y, t, \widetilde{Re})$, dimension 4. Outputs: (u, v, p), dimension 3. Total trainable parameters: $4 \times 64 + 5 \times (64 \times 64 + 64) + 64 \times 3 + 3 = 21\,379$.

Control network. Four hidden layers, each with 32 neurons, using tanh activation. Inputs: $(\widetilde{Re})$, dimension 1. Outputs: $\{a_{mn}\}_{m=1,\ldots,N_x;\, n=0,\ldots,N_t-1}$, dimension $N_x \times N_t = 30$ (baseline). Total trainable parameters: $1 \times 32 + 4 \times (32 \times 32 + 32) + 32 \times 30 + 30 = 3\,614$.

Initialization. All network weights are initialized using the Xavier/Glorot normal distribution with zero-mean and variance $\sigma^2 = 2/(n_{\text{in}} + n_{\text{out}})$ where n_{in} and n_{out} are the fan-in and fan-out of each layer. All biases are initialized to zero.

Precision. All weight tensors, input tensors, automatic differentiation operations, and loss evaluations use 64-bit (double) precision. This choice avoids numerical cancellation in the evaluation of the PDE residuals, which involve second-order derivatives of the network outputs.

Table S1 summarizes the architecture specifications.

1.2 S1.2 Collocation Sampling and Initialization

All collocation points are resampled from scratch at the beginning of each training epoch, preventing overfitting to any fixed quadrature set.

Table S1: Neural network architecture specifications for flow and control networks.

Network	Layers	Neurons	Activation	Inputs	Outputs
Flow	6 hidden + I/O	64	tanh	4	3 (u,v,p)
Control	4 hidden + I/O	32	tanh	1	30 (a_{mn})

Interior PDE collocation: $N_{\text{phys}} = 4\,000$ points sampled uniformly at random over the computational domain $\Omega \times [0,1] = [0, L_x] \times [0,1] \times [0,1]$. At each epoch, three Reynolds-number values are selected from the set $\{100, 300, 500, 700, 900, 1000\}$; each interior point is evaluated at all three selected Re values, giving an effective PDE sample size of $3N_{\text{phys}} = 12\,000$ residual evaluations per epoch.

Boundary collocation: $N_{\text{BC}} = 600$ points distributed across the boundary $\partial\Omega$, split as 200 on Γ_{top} (upper lid) and 400 on Γ_{stat} (bottom and lateral walls), resampled uniformly at each epoch.

Reynolds parameter: During each epoch, 6 discrete Reynolds values (100, 300, 500, 700, 900, 1000) are used for boundary-condition enforcement and the Reynolds-separation regularization.

1.3 S1.3 Optimization Schedule and Hyperparameters

Training proceeds in two stages:

Stage 1 — Pre-training (300 epochs): Only the flow network weights are updated. The control coefficients a_{mn} are frozen at zero, preventing the optimizer from immediately collapsing to degenerate control solutions before a physically consistent flow representation has been established. During this stage, only the PDE residual, boundary-condition, and Reynolds-separation losses are active (the dissipation and energy objectives depend on flow accuracy to be meaningful).

Stage 2 — Joint optimization (3 000 epochs): All flow and control network weights are updated simultaneously. The full multi-objective loss (Main Eq. (13)) is active with the reference weights listed after that equation.

Optimizer: Adam with $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon_{\text{Adam}} = 10^{-8}$.

Learning rate schedule: Initial rate $\eta_0 = 8 \times 10^{-4}$ with exponential decay by factor 0.995 every 100 epochs, giving $\eta(e) = \eta_0 \times 0.995^{\lfloor e/100 \rfloor}$. After 3 000 epochs, the effective learning rate is approximately $8 \times 10^{-4} \times 0.995^{30} \approx 6.9 \times 10^{-4}$, so the learning rate has decayed by only $\sim 14\%$ over the full training. No gradient clipping is applied.

Convergence behavior: Typical behavior over the 3 000-epoch joint phase shows $\mathcal{L}_{\text{PDE}}$ reaching approximately 10^{-3} by epoch 3 000, $\mathcal{L}_{\text{BC}}$ below 5×10^{-4}, and $\mathcal{L}_{\text{ctrl}}$ below 10^{-4} (corresponding to $|E_r - E^*| < 10^{-2}$). The total loss value at convergence lies in the range $[1.37, 1.40]$ for the $(Re, E^*) = (500, 0.25)$ condition used in the seed study.

Seed sensitivity: The early training stochasticity during Stage 1 determines which outcome cluster is entered during Stage 2. Once Stage 2 begins, the optimizer rarely switches clusters, explaining the reproducible classification of each realization. No early stopping is applied; all realizations run for the full 3 000 epochs regardless of convergence.

1.4 S1.4 Multi-Objective Loss Formulation

The complete loss is (Main Eq. (13)):

$$\mathcal{L} = \lambda_{\text{PDE}}\mathcal{L}_{\text{PDE}} + \lambda_{\text{BC}}\mathcal{L}_{\text{BC}} + \lambda_{\text{diss}}\mathcal{L}_{\text{diss}} + \lambda_{\text{ctrl}}\mathcal{L}_{\text{ctrl}} + \lambda_{\text{var}}\mathcal{L}_{\text{var}} + \lambda_{Re}\mathcal{L}_{Re}, \tag{2}$$

with baseline weights $\lambda_{\mathrm{PDE}} = 1$, $\lambda_{\mathrm{BC}} = 15$, $\lambda_{\mathrm{diss}} = 0.05$, $\lambda_{\mathrm{ctrl}} = 200$, $\lambda_{\mathrm{var}} = 1$, $\lambda_{Re} = 2$. The individual components are:

$$\mathcal{L}_{\mathrm{PDE}} = \left\langle R_u^2 + R_v^2 + R_c^2 \right\rangle_{\mathcal{X}_{\mathrm{int}}}, \tag{3}$$

$$\mathcal{L}_{\mathrm{BC}} = \left\langle u^2 + v^2 \right\rangle_{\Gamma_{\mathrm{stat}}} + \left\langle (u - U_{\mathrm{lid}})^2 + v^2 \right\rangle_{\Gamma_{\mathrm{top}}}, \tag{4}$$

$$\mathcal{L}_{\mathrm{diss}} = \int_\Omega \frac{1}{Re}\left[u_x^2 + v_y^2 + \tfrac{1}{2}(u_y + v_x)^2\right] d\Omega = \int_\Omega \nu\, S{:}S \, d\Omega, \tag{5}$$

$$\mathcal{L}_{\mathrm{ctrl}} = (E_r - E^*)^2, \tag{6}$$

$$\mathcal{L}_{\mathrm{var}} = -\log\left[\mathrm{Var}(U_{\mathrm{lid}}) + 10^{-5}\right], \tag{7}$$

$$\mathcal{L}_{Re} = \max\left[0,\ \delta - \frac{1}{N_{\mathrm{probe}}} \sum_{i=1}^{N_{\mathrm{probe}}} \left(U_{\mathrm{lid}}(x_i, t_0, Re_{\min}) - U_{\mathrm{lid}}(x_i, t_0, Re_{\max})\right)^2\right], \tag{8}$$

with $N_{\mathrm{probe}} = 20$ probe locations distributed uniformly in $x \in [0, L_x]$, $t_0 = 0.5$, $\delta = 0.01$, $Re_{\min} = 100$, $Re_{\max} = 1000$.

No initial-condition term is included by design. The temporal Fourier expansion and the cycle-averaged objective define a periodic space–time boundary-value problem, so training targets a statistically stationary periodic state rather than the transient from a prescribed initial field (Main Sec. 4.1).

Weight rationale. The weight $\lambda_{\mathrm{BC}} = 15$ strongly enforces boundary conditions relative to the interior residual, preventing poorly conditioned flow states where boundary violations dominate the solution. $\lambda_{\mathrm{ctrl}} = 200$ is large to enforce the energy constraint tightly ($|E_r - E^*|/E^* < 5\%$ at convergence for all tested conditions), ensuring that all realizations are compared at effectively the same actuation energy. The small value $\lambda_{\mathrm{diss}} = 0.05$ prevents the dissipation objective from overwhelming the PDE constraints during early training, which can cause the optimizer to ignore physical accuracy in favor of reducing dissipation. The regularizer 10^{-5} in $\mathcal{L}_{\mathrm{var}}$ prevents numerical issues when $\mathrm{Var}(U_{\mathrm{lid}}) = 0$ at the beginning of training before the control network has been activated.

2 S2. Control-Space and Modal-Analysis Details

2.1 S2.1 Spatio-Temporal Fourier Basis

The boundary actuation is expanded as in Main Sec. 3.2:

$$U_{\mathrm{lid}}(x, t, Re) = \sum_{m=1}^{N_x} \sum_{n=0}^{N_t - 1} a_{mn}(Re) \sin\left(\frac{m\pi x}{L_x}\right) \cos(n\pi t), \tag{9}$$

with baseline truncation $N_x = 6$, $N_t = 5$, giving 30 nominal coefficients. The sine basis $\sin(m\pi x/L_x)$ is chosen because it vanishes at $x = 0$ and $x = L_x$ for all $m \geq 1$, automatically enforcing corner compatibility without any additional constraint. The cosine basis $\cos(n\pi t)$ has the property that $\cos(0 \cdot \pi t) = 1$, so the $n = 0$ terms represent purely stationary (time-independent) actuation. All mode indices in this manuscript use the physical (one-based) convention: the second spatial harmonic corresponds to $m = 2$, not $m = 1$.

2.2 S2.2 Modal Energy Normalization and Quadrature

The time-average of the squared basis functions satisfies $\langle \cos^2(0) \rangle_t = 1$ and $\langle \cos^2(n\pi t) \rangle_t = \frac{1}{2}$ for $n \geq 1$. The space-average of the squared sine basis functions satisfies $\langle \sin^2(m\pi s) \rangle_s = \frac{1}{2}$ for all $m \geq 1$, where $s = x/L_x \in [0, 1]$. Therefore, the contribution of coefficient a_{mn} to the realized

energy $E_r = \langle U_{\mathrm{lid}}^2 \rangle_{s,t}$ is:

$$E_{mn} = \langle a_{mn}^2 \sin^2(m\pi s)\cos^2(n\pi t)\rangle_{s,t} = \begin{cases} \frac{1}{2}a_{m0}^2 & n = 0, \\ \frac{1}{4}a_{mn}^2 & n \geq 1. \end{cases} \tag{10}$$

These are the modal energies $\mathcal{E}_{mn}$ used in Main Sec. 3.2. The realized actuation energy is therefore

$$E_r = \sum_{m=1}^{N_x} \left(\tfrac{1}{2}a_{m0}^2 + \sum_{n=1}^{N_t-1} \tfrac{1}{4}a_{mn}^2 \right). \tag{11}$$

2.3 S2.3 Branch Classification Thresholds

The branch clusters are defined by the following explicit rule applied in order to the normalized modal fractions $f_1 \equiv f_{10}$ (stationary first spatial harmonic), $f_2 \equiv f_{20}$ and $f_t = \sum_{m,n\geq 1} f_{mn}$:

- B_{first}: $f_1 > 0.5$ (stationary first harmonic carries an absolute majority of the actuation energy).
- B_1: $f_2 > 2f_t$ and $f_2 > 0.5$ (stationary second mode carries at least twice the energy of all temporal modes, and more than half of the total).
- B_2: $f_t > 2f_2$ and $f_t > 0.5$ (temporal family dominant).
- B_3: none of the above (residual class).

This is a dominance rule rather than an absolute cut-off on f_2 or f_t: a realization is assigned to the class that carries the control energy, and the rule reproduces the three well-separated groups visible in the (f_2, f_t) observable space of the identification campaign. The parameters $k = 2$ (dominance factor) and $\phi = 0.5$ (floor) were fixed a priori and not tuned to the data.

The B_{first} test was added after a full-spectrum audit of all 185 realizations and of the λ_{var} campaign, because the earlier three-class rule assigned a first-harmonic-dominated control to B_3 whenever it satisfied neither $f_2 > 0.5$ nor $f_t > 0.5$. The audit shows that this defect never materializes in the broad campaign: $f_1 \leq 0.122$ over all 185 runs ($f_1 \leq 0.096$ within B_3), so B_{first} is empty there and all 185 labels, frequencies and confidence intervals are unchanged (the class audit of Sec. S2.3). It does materialize in the ablation campaign, where removing $\mathcal{L}_{\mathrm{var}}$ produces $f_1 = 0.87$–0.99 (Table S7). The audit also resolves what B_3 contains: 15 of its 26 members are dominated by the temporal family, 8 by the stationary second harmonic and 3 by stationary harmonics of order $m \geq 3$; none by the first harmonic (sub-structure analysis of Sec. S2.3).

Table S2 reports the sensitivity of the pooled frequencies to both k and ϕ: varying k between 1.5 and 3, or ϕ between 0.4 and 0.6, moves $\widehat{P}(B_1)$ and $\widehat{P}(B_2)$ by less than three percentage points and changes at most 12 of the 185 realizations. The frequency of the residual class B_3 is more sensitive (9.2%–20.5%) because it is defined as the complement of the other conditions and is the smallest class; its share should therefore be read as approximate, whereas the B_1/B_2 balance used in this study is robust. Lowering the B_{first} threshold from 0.5 to 0.2 or even 0.13 would still leave the class empty in this campaign, so the pooled frequencies are insensitive to that choice. Only 6 realizations lie within 0.05 of a dominance boundary, and the smallest distance to a boundary over the whole campaign is 0.008.

3 S3. 185-Realization Campaign: Cell-Aggregated Dataset

The broad (Re, E^*) campaign evaluated 185 independent training runs across 30 cells in the parameter domain ($Re \in \{100, 300, 500, 700, 900, 1000\}$, $E^* \in \{0.1, 0.25, 0.5, 1, 2\}$). The baseline

Table S2: Sensitivity of the pooled branch frequencies to the parameters of the dominance rule (B_1: $f_2 > kf_t$ and $f_2 > \phi$; B_2: $f_t > kf_2$ and $f_t > \phi$; B_3: otherwise). The reference rule is $k = 2$, $\phi = 0.5$; one parameter is varied at a time. The last column counts the realizations whose label changes with respect to the reference.

Parameter	Value	B_1 (%)	B_2 (%)	B_3 (%)	Reclassified (of 185)
dominance factor k	1.50	43.2	47.6	9.2	9
dominance factor k	1.75	42.7	45.4	11.9	4
dominance factor k	2.00 (reference)	42.2	43.8	14.1	0
dominance factor k	2.50	40.5	41.6	17.8	7
dominance factor k	3.00	39.5	40.0	20.5	12
floor ϕ	0.40	42.2	44.9	13.0	2
floor ϕ	0.50 (reference)	42.2	43.8	14.1	0
floor ϕ	0.60	41.1	40.5	18.4	8

cell size is $N_s = 5$ seeds per cell; selected cells of particular interest received 10, 15, or 20 seeds. The global aggregate counts are B_1: 78 (42.2%), B_2: 81 (43.8%), B_3: 26 (14.1%). Table S3 reports aggregate metrics for all 30 cells.

Table S3: Cell-aggregated statistics for the 30 cells of the broad campaign. $\widehat{P}(B_k)$ are empirical recovery frequencies of branch clusters. H_N is the normalized branch-selection entropy. Small sample sizes per cell limit precision of frequency estimates (see Sec. S4.2).

Cell ID	Re	E^*	N_s	$\widehat{P}(B_1)$	$\widehat{P}(B_2)$	$\widehat{P}(B_3)$	H_N	Dominant
re100_e0.1	100	0.10	5	0.00	0.80	0.20	0.455	B_2
re300_e0.1	300	0.10	5	0.40	0.60	0.00	0.613	B_2
re500_e0.1	500	0.10	5	0.80	0.20	0.00	0.455	B_1
re700_e0.1	700	0.10	5	0.60	0.40	0.00	0.613	B_1
re900_e0.1	900	0.10	5	0.60	0.40	0.00	0.613	B_1
re1000_e0.1	1000	0.10	5	0.60	0.40	0.00	0.613	B_1
re100_e0.25	100	0.25	5	0.00	0.80	0.20	0.455	B_2
re300_e0.25	300	0.25	5	0.20	0.60	0.20	0.865	B_2
re500_e0.25	500	0.25	15	0.467	0.400	0.133	0.902	B_1
re700_e0.25	700	0.25	5	0.40	0.40	0.20	0.960	B_1/B_2
re900_e0.25	900	0.25	5	0.40	0.40	0.20	0.960	B_1/B_2
re1000_e0.25	1000	0.25	5	0.40	0.40	0.20	0.960	B_1/B_2
re100_e0.5	100	0.50	5	0.40	0.60	0.00	0.613	B_2
re300_e0.5	300	0.50	5	0.40	0.60	0.00	0.613	B_2
re500_e0.5	500	0.50	5	0.40	0.60	0.00	0.613	B_2
re700_e0.5	700	0.50	5	0.40	0.40	0.20	0.960	B_1/B_2
re900_e0.5	900	0.50	5	0.40	0.40	0.20	0.960	B_1/B_2
re1000_e0.5	1000	0.50	5	0.40	0.40	0.20	0.960	B_1/B_2
re100_e1	100	1.00	5	0.40	0.60	0.00	0.613	B_2
re300_e1	300	1.00	5	0.40	0.40	0.20	0.960	B_1/B_2
re500_e1	500	1.00	5	0.60	0.40	0.00	0.613	B_1
re700_e1	700	1.00	5	0.60	0.40	0.00	0.613	B_1
re900_e1	900	1.00	5	0.40	0.40	0.20	0.960	B_1/B_2

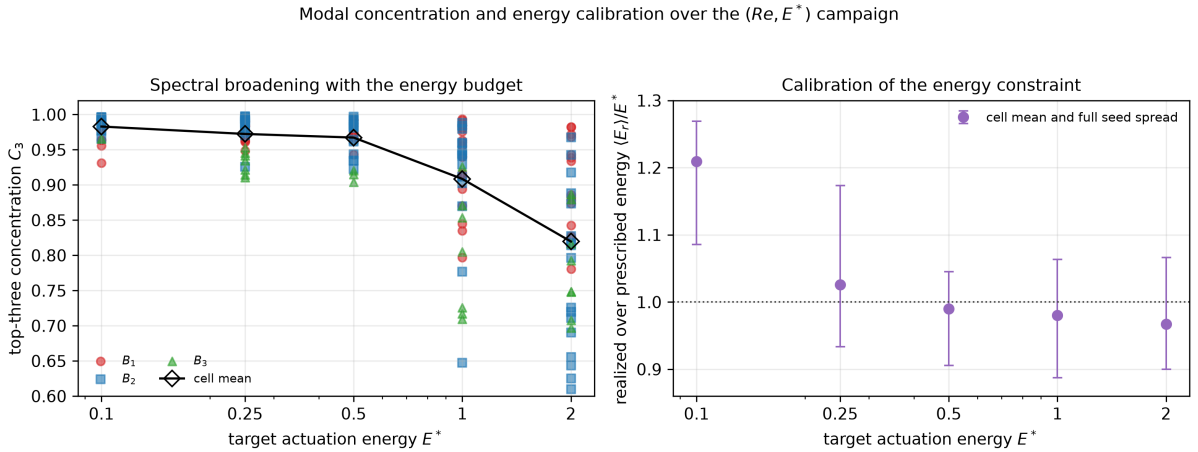


Figure S1: Modal concentration and energy calibration over the 185 realizations of the broad (Re, E^*) campaign. Left: top-three spectral concentration C_3 of the lid control for every realization, colored by assigned branch class, with the cell mean. Right: realized over prescribed actuation energy, cell mean with the full seed spread. The dotted line is the nominal value $E_r = E^*$.

(Continued)

Cell ID	Re	E^*	N_s	$\widehat{P}(B_1)$	$\widehat{P}(B_2)$	$\widehat{P}(B_3)$	H_N	Dominant
re1000_e1	1000	1.00	20	0.550	0.200	0.250	0.908	B_1
re100_e2	100	2.00	5	0.60	0.40	0.00	0.613	B_1
re300_e2	300	2.00	5	0.40	0.60	0.00	0.613	B_2
re500_e2	500	2.00	5	0.40	0.60	0.00	0.613	B_2
re700_e2	700	2.00	15	0.200	0.400	0.400	0.960	B_2/B_3
re900_e2	900	2.00	5	0.40	0.40	0.20	0.960	B_1/B_2
re1000_e2	1000	2.00	5	0.40	0.40	0.20	0.960	B_1/B_2

The B_1 branch is consistently accessible at moderate Reynolds numbers ($Re = 500$–700) for low-to-moderate actuation energies, and at high $Re = 1000$ it has the highest accessibility frequency at $E^* = 1$ (55%). The B_2 branch dominates at low Re (100–300), consistent with the observation that temporal actuation patterns are more easily accessible when the flow is highly viscous and the second-harmonic outcome is less frequently recovered. The B_3 branch appears primarily at high actuation energies and moderate-to-high Reynolds numbers.

Two properties of the dataset condition every frequency quoted in the main text. Figure S1 shows them. First, the top-three modal concentration C_3, the sum of the three largest energy fractions of the lid control, decreases monotonically from 0.983 at $E^* = 0.1$ to 0.909 at $E^* = 1$ and 0.820 at $E^* = 2$: raising the energy budget spreads the control over more harmonics, which is why the two dominant classes are sharply separated at low energy and why the residual class B_3 is populated mainly at high energy. Second, the realized energy is not exactly the prescribed one. The cell mean of $\langle E_r \rangle / E^*$ lies within 3% of unity for every $E^* \geq 0.25$, but at $E^* = 0.1$ all thirty realizations of that column overshoot, with a mean of 1.209 and a spread of 1.086–1.269. The energy penalty is therefore looser in the low-budget regime than elsewhere, and the $E^* = 0.1$ column of the map is explored at an actuation energy about 20% larger than its label. This is a property of the soft penalty formulation, not of the solver, and it does not affect the iso-energetic comparisons of Sec. S8, which rescale each control to a common realized energy before the lattice-Boltzmann transfer.

4 S4. Complete λ_{var} Campaign and Confidence Intervals

4.1 S4.1 Campaign Design

The variance-regularization campaign comprised 90 realizations distributed across 18 parameter combinations: 3 operating points ($(Re, E^*) \in \{(500, 0.25), (700, 2), (1000, 1)\}$) times 6 values of $\lambda_{\text{var}} \in \{0, 0.25, 0.5, 1, 2, 5\}$, with $N_s = 5$ independent random seeds per combination. All other training hyperparameters are held fixed at the reference values given in Sec. S1.

4.2 S4.2 Binomial Wilson Score Confidence Intervals

For $N_s = 5$ trials, the 95% Wilson score confidence intervals [lo, hi] for the empirical branch-selection frequency $\widehat{P} = k/5$ are computed using the formula

$$\left[\frac{\hat{p} + \frac{z^2}{2N} \mp z\sqrt{\frac{\hat{p}(1-\hat{p})}{N} + \frac{z^2}{4N^2}}}{1 + z^2/N} \right], \tag{12}$$

with $z = 1.96$ (95% coverage) and $N = 5$. The resulting intervals are reported in Table S4.

Table S4: 95% Wilson score confidence intervals for empirical branch-selection frequencies at $N_s = 5$.

k (successes)	$\widehat{P} = k/5$	95% CI [lo, hi]
0	0.00	[0.000, 0.434]
1	0.20	[0.036, 0.624]
2	0.40	[0.118, 0.769]
3	0.60	[0.231, 0.882]
4	0.80	[0.376, 0.964]
5	1.00	[0.566, 1.000]

These wide intervals reflect the small sample size ($N_s = 5$) and should be interpreted accordingly. The main conclusions of the λ_{var} campaign rest on the qualitative trends (e.g., all conditions yielding exclusively B_{first} at $\lambda_{\text{var}} = 0$, consistent across all 3 operating points) rather than on precise frequency estimates at individual cells. The classification of those zero-weight controls is itself part of the audit of Sec. S2.3: they are first-harmonic dominated, not the residual B_3 class that the earlier three-class rule assigned them to. The sensitivity of the classification itself is analysed in Sec. S2.3 (Table S2): the two dominant classes are stable to within three percentage points, while the share of the intermediate class B_3 is accurate only to about ± 5 points.

4.3 S4.3 Interpretation of Non-Monotonic Behavior

The non-monotonic dependence of $\widehat{P}(B_1)$ on λ_{var} observed at $(Re, E^*) = (500, 0.25)$ (maximal at intermediate values, decreasing at large values) reflects the competing effects of variance regularization on the two structured branches. At small λ_{var}, the variance penalty provides a weak preference for spatially structured controls, leading to roughly equal B_1 and B_2 accessibility. At large λ_{var}, the variance penalty strongly rewards any spatially structured profile, and the time-dependent B_2 branch (which generally achieves higher spatial variance due to its temporal oscillations producing an effective spatial pattern) becomes relatively more accessible. This operating-point-dependent response explains why the optimal λ_{var} for B_1 selection differs between conditions.

5 S5. Energy, Initialization, and Loss-Ablation Studies

5.1 S5.2 Detailed Ten-Seed Experiment at $Re = 500$, $E^* = 0.25$

Table S5 reports the complete seed-level metrics for the 10-seed multiplicity study. The key result is the tight loss range $[1.371, 1.394]$ across all seeds despite qualitatively distinct modal structures. Seed 9 achieves the lowest loss (1.371) and is classified as B_2, while seed 4 achieves the highest loss (1.394) and is classified as B_3. The four B_1 realizations span losses from 1.372 to 1.377, interleaved with the four B_2 realizations that span 1.371 to 1.377. This interleaving provides direct evidence that the loss landscape does not have a clear ordering between the two branch types.

Table S5: Seed-level metrics for the ten-seed experiment at $Re = 500$, $E^* = 0.25$.

Seed	E_r	f_2 (%)	f_t (%)	Total loss	Classification
0	0.240	53.2	43.8	1.377	Intermediate (B_3)
1	0.267	91.1	1.3	1.377	Second-mode (B_1)
2	0.257	89.2	5.0	1.374	Second-mode (B_1)
3	0.268	2.1	97.8	1.372	Temporal (B_2)
4	0.250	52.2	45.0	1.394	Intermediate (B_3)
5	0.245	0.4	99.5	1.373	Temporal (B_2)
6	0.265	86.3	1.0	1.376	Second-mode (B_1)
7	0.250	91.3	1.6	1.374	Second-mode (B_1)
8	0.250	1.3	98.4	1.372	Temporal (B_2)
9	0.267	0.1	99.6	1.371	Temporal (B_2)

5.2 S5.3 Extended Loss-Term Ablation

Table S6 summarizes all ablation configurations, including dual-term removals and modified weight values. The single most important result is the complete modal-class shift when $\mathcal{L}_{\text{var}}$ is removed. The dual ablation results (removing both $\mathcal{L}_{\text{var}}$ and one other term) confirm that the shift is driven entirely by the absence of $\mathcal{L}_{\text{var}}$, not by any interaction with the removed secondary term. Reducing λ_{var} to half its reference value causes a modest 3% decrease in f_2, while doubling it increases f_2 by 0.6%, suggesting a nearly saturated response at the reference weight $\lambda_{\text{var}} = 1$.

Table S6: Complete loss-term ablation at $Re = 500$, $E^* = 0.25$. Each entry is a single converged realization with the specified configuration and the same random seed as the reference Baseline.

Configuration	E_r	f_2 (%)	f_t (%)	Interpretation
Baseline	0.2534	92.2	1.3	Second-mode dominated
No $\mathcal{L}_{\text{var}}$	0.2582	0.2	5.1	Shift to B_{first}
No $\mathcal{L}_{Re}$	0.2534	92.2	1.3	Essentially unchanged
No $\mathcal{L}_{\text{diss}}$	0.2533	92.3	1.2	Essentially unchanged
No $\mathcal{L}_{\text{var}}, \mathcal{L}_{Re}$	0.2582	0.2	5.1	Same first-harmonic shift
No $\mathcal{L}_{\text{var}}, \mathcal{L}_{\text{diss}}$	0.2578	0.3	5.0	Same first-harmonic shift
$\lambda_{\text{var}}/2$	0.2546	89.3	1.5	Reduced concentration
$2\lambda_{\text{var}}$	0.2598	92.8	1.0	Slightly increased

Figure S2 makes the two independent statements of the ablation visible. The realized energy is essentially unaffected by every configuration (0.2533–0.2598, all within 4% of $E^* = 0.25$), so the configurations are comparable at fixed budget; what changes is only the distribution of that budget between the harmonic families. The three $\mathcal{L}_{\text{var}}$ removals collapse f_2 from 92.2% to

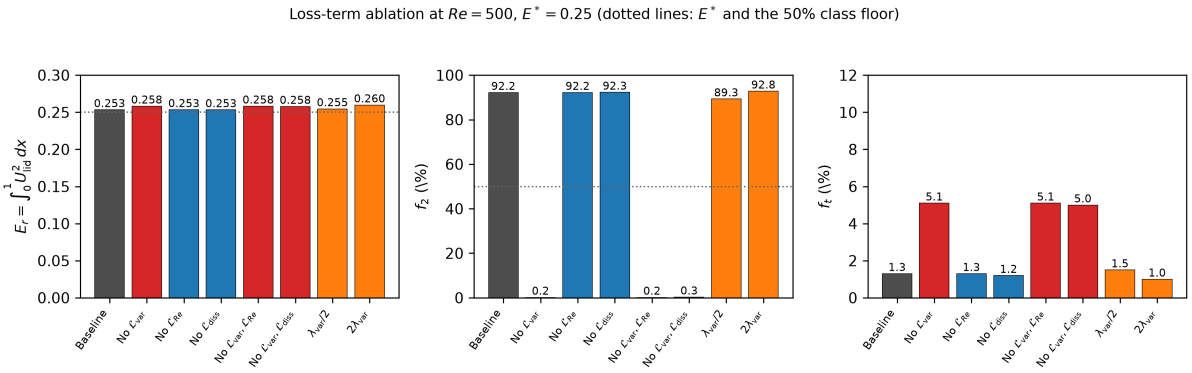


Figure S2: The eight configurations of Table S6 in graphical form: realized actuation energy, stationary second-harmonic fraction and total time-dependent fraction. Grey: baseline; red: the three configurations that remove $\mathcal{L}_{\mathrm{var}}$; blue: the single removals of the other two terms; orange: the two rescalings of λ_{var}. Only the red configurations change the outcome class, and all eight keep E_r within 4% of the target.

Table S7: First-harmonic content of the $\lambda_{\mathrm{var}} = 0$ controls (five seeds per operating point, archived in `results/02_lambda_var`). Every realization is dominated by the first spatial harmonic and is reclassified as B_{first}; the second-harmonic fraction never exceeds 2.5%.

Operating point	E_r target	f_1 (%)	f_2 (%)	f_t (%)	B_{first} seeds
$Re = 1000$, $E^* = 1$	1	86.8–97.2	≤ 1.02	≤ 12.1	5/5
$Re = 500$, $E^* = 0.25$	0.25	89.9–93.6	≤ 0.31	≤ 5.4	5/5
$Re = 700$, $E^* = 2$	2	93.1–98.6	≤ 2.41	≤ 4.5	5/5

0.2–0.3% and raise f_t from 1.3% to 5.0–5.1%, while the two other single-term removals leave both fractions unchanged to within 0.1 percentage points. Note that the ablated runs are not B_2 either, and they are not the residual B_3: with $f_2 = 0.2\%$ and $f_t = 5.1\%$ the remaining energy sits in the stationary first harmonic, so the outcome is the first-harmonic class B_{first} of the extended rule (Main Sec. 3.2). The archived five-seed repetition of this configuration confirms the point: $f_1 = 87$–99%, $f_2 \leq 2.5\%$, $f_t \leq 12\%$ at all three operating points (Table S7). The earlier label B_3 for these runs was therefore a naming artifact of a rule that did not test the first harmonic.

Table S8 decomposes the converged loss into its weighted contributions for one representative realization of each branch type. The three total losses agree to better than 0.3%, but the individual terms differ by up to a factor of 2.4 in the PDE contribution. In particular, the weighted variance term is essentially identical for all three branches ($\lambda_{\mathrm{var}}\mathcal{L}_{\mathrm{var}} \approx 0.2595$ in each case), so the variance penalty does not order the converged solutions; its role is to shape the path taken during training, consistent with the complete modal-class shift observed when it is removed. Differences between branches are therefore concentrated in the physical residuals (PDE, boundary condition and dissipation terms at the 10^{-3} level), and similar total losses must not be read as similar physical performance.

6 S6. Network Capacity and Control-Truncation Studies

6.1 S6.1 Network Capacity Sensitivity

Three network configurations were evaluated at $Re = 500$, $E^* = 0.25$; they are listed in Table S9.

The f_2 values span only $[89.5\%, 92.2\%]$, a range of 2.7% across a 40-fold increase in network parameters. This robustness suggests that the second-harmonic concentration is not imposed

Table S8: Decomposition of the converged weighted loss terms for one representative realization of each branch type at $(Re, E^*) = (500, 0.25)$. Similar total losses do not imply comparable physical objectives.

Branch (seed)	$\lambda_{\text{PDE}}\mathcal{L}_{\text{PDE}}$	$\lambda_{\text{BC}}\mathcal{L}_{\text{BC}}$	$\lambda_{\text{diss}}\mathcal{L}_{\text{diss}}$	$\lambda_{\text{ctrl}}\mathcal{L}_{\text{ctrl}}$	$\lambda_{\text{var}}\mathcal{L}_{\text{var}}$	$\mathcal{L}_{\text{total}}$	E_r	$\lvert E_r - E^*\rvert^2$
B1 (7)	5.388e-03	1.438e-03	2.376e-04	1.802e-02	2.595e-01	1.3741	0.2595	9.008e-05
B2 (3)	2.901e-03	2.084e-03	5.149e-04	1.824e-02	2.595e-01	1.3725	0.2595	9.119e-05
B3 (0)	6.936e-03	2.586e-03	2.491e-04	1.797e-02	2.595e-01	1.3768	0.2595	8.985e-05

Table S9: Network capacity sensitivity at $Re = 500$, $E^* = 0.25$.

Configuration	Layers	Neurons	Parameters	E_r	f_2 (%)	f_t (%)
Small	4	32	≈4 200	0.2541	91.8	1.2
Reference	6	64	21 379	0.2534	92.2	1.3
Large	8	128	≈170 000	0.2518	89.5	2.1

by the inductive bias of the network architecture but reflects a genuine optimization landscape feature. The large network shows a 2.7% decrease in f_2 relative to the reference, which may reflect that increased expressivity allows the network to slightly distribute energy into higher harmonics that are otherwise suppressed by the effective regularization of limited parameter count.

6.2 S6.2 Control-Space Truncation Details

Six control basis truncations were evaluated at $Re = 500$, $E^* = 0.25$; the complete results are given in Table S10.

Table S10: Complete control-truncation sensitivity results at $Re = 500$, $E^* = 0.25$.

(N_x, N_t)	Modes	A_2	E_r	f_2 (%)	f_t (%)	Class
(4,3)	12	0.671	0.2541	87.6	2.1	B_1
(6,1)	6	0.678	0.2533	89.2	0.0	B_1
(6,3)	18	0.678	0.2778	89.3	1.8	B_1
(6,5)	30	0.684	0.2533	92.3	1.3	B_1 (reference)
(8,5)	40	0.669	0.2475	87.8	2.0	B_1
(8,9)	72	0.668	0.2775	87.3	2.4	B_1

The $(6, 1)$ truncation (stationary modes only) achieves $f_2 = 89.2\%$ and $f_t = 0.0\%$ by construction: with no temporal modes available, the optimizer can only select stationary controls and the second harmonic is the dominant choice. The close f_2 values between $(6, 1)$ and the other truncations suggest that temporal modes are not strongly favored by the optimizer when Fourier basis modes are available, confirming the intrinsic preference for the stationary second-harmonic structure. The somewhat lower f_2 at the $(4, 3)$ and $(8, x)$ bases may reflect the different Fourier mode structures available in the truncated space.

7 S7. Geometry and Control-Parametrization Studies

7.1 S7.1 Cavity Aspect Ratio Details

The aspect-ratio sensitivity study evaluates three geometries at fixed $Re = 500$, $E^* = 0.25$, listed in Table S11.

For $AR = 0.5$, the second harmonic of the Fourier basis is $\sin(2\pi x/0.5) = \sin(4\pi x)$, which has a half-wavelength of 0.25—much shorter than the cavity height. This shorter wavelength may

Table S11: Aspect-ratio sensitivity at $Re = 500$, $E^* = 0.25$.

AR	L_x	A_2	E_r	f_2 (%)	f_t (%)
0.5	0.5	0.6034	0.2603	69.9	26.7
1.0	1.0	0.6835	0.2534	92.2	1.3
2.0	2.0	0.6770	0.2514	91.2	1.6

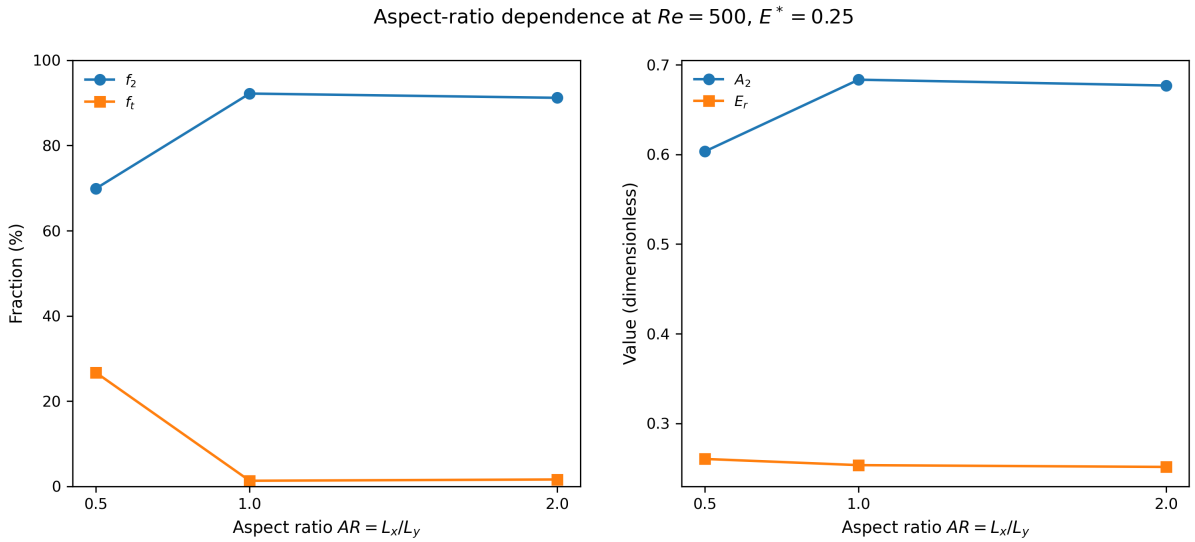


Figure S3: Aspect-ratio dependence of the modal structure at $Re = 500$, $E^* = 0.25$: second-harmonic and temporal fractions (left) and second-mode amplitude and realized energy (right), for the three geometries of the table above. The realized energy stays within 4.1% of the target at every aspect ratio, so the drop of f_2 at $AR = 0.5$ is a redistribution of the same budget, not a change of budget.

be less efficient at redistributing the primary cavity vortex, explaining the reduced $f_2 = 69.9\%$. For $AR = 2.0$, the second harmonic is $\sin(2\pi x/2) = \sin(\pi x)$, which has a half-wavelength of 1—equal to the cavity height, potentially generating a more favorable symmetric structure, consistent with $f_2 = 91.2\%$.

Figure S3 shows the same data together with the two quantities that keep the comparison meaningful: E_r varies from 0.2514 to 0.2603, i.e. by less than 3.5% over the three geometries, while A_2 falls from 0.6835 to 0.6034 at $AR = 0.5$ and f_t rises from 1.3% to 26.7%. The compressed geometry therefore does not receive a larger budget; it spends a quarter of the same budget on time-dependent coefficients.

7.2 S7.2 Modified-Chebyshev Basis Construction

The modified-Chebyshev parameterization employs normalized Chebyshev polynomials $T_k(\xi)$ evaluated at $\xi = 2x/L_x - 1 \in [-1, 1]$, modified by the factor $(1 - \xi^2)$ to enforce zero velocity at both lateral walls:

$$\psi_k(x) = (1 - \xi^2)T_k(\xi), \quad k = 0, 1, 2, \ldots \tag{13}$$

The first four basis functions are:

$$\psi_0(x) = (1-\xi^2), \tag{14}$$
$$\psi_1(x) = (1-\xi^2)\xi, \tag{15}$$
$$\psi_2(x) = (1-\xi^2)(2\xi^2-1), \tag{16}$$
$$\psi_3(x) = (1-\xi^2)(4\xi^3-3\xi). \tag{17}$$

These functions are smooth, vanish at $x = 0$ and $x = L_x$ for all k, and provide a dense covering of the control space. However, unlike the Fourier sine basis, they are not orthogonal in the $L^2[0, L_x]$ inner product, so modal energies computed in this basis are not directly comparable to the Fourier fractions f_2 and f_t.

7.3 S7.3 Basis Reconstruction Accuracy

The reconstruction error of the reference Fourier B_1 control profile by the Chebyshev basis is measured as the RMSE between the target profile $U_{\text{lid,Fourier}}$ and its best approximation in the Chebyshev basis using the same number of coefficients. The Fourier basis achieves machine-precision reconstruction by construction (RMSE $= 1.6 \times 10^{-16}$), while the Chebyshev basis achieves RMSE $= 3.8 \times 10^{-3}$, confirming that the two bases represent qualitatively different function classes despite both satisfying the lateral-wall zero conditions.

7.4 S7.4 Quasi-Stationarity as a Representation Diagnostic

The quasi-stationarity results reported in this section demonstrate that the representation choice determines not only the Fourier modal structure but the actual physical flow state. Fourier B_1 controls produce $R_K \leq 0.65\%$ at all tested Re, consistent with a quasi-static flow response to the time-averaged lid profile. The Chebyshev controls, which have $f_t > 93\%$, impose strongly oscillatory boundary conditions that drive persistent unsteady flows with $R_K \geq 44\%$. This physical difference is quantitative, not merely terminological: when the Chebyshev flow is averaged over its oscillation period, its dissipation is 2.0×10^{-2} to 1.5×10^{-1}, i.e. the *largest* of all configurations examined, whereas the dissipation of its time-mean field (2.4×10^{-4} to 6.0×10^{-4}) is the smallest (Sec. S8.5). A comparison based on the mean field therefore reverses the ranking, which is why the two representations cannot be ranked on $\varepsilon(\langle u \rangle)$ and why the temporal controls are reported as cycle averages. The LBM dissipation comparison of the main text (Main Table 2) is reported only for the quasi-stationary Fourier controls, for which a single steady-state dissipation value is well defined.

8 S8. Independent MRT-LBM Verification and Grid Uncertainty

8.1 S8.1 MRT-LBM Formulation

The independent lattice-Boltzmann solver uses the D2Q9 model with 9 velocity directions $\mathbf{c}_i$ ($i = 0, \ldots, 8$) defined as $\mathbf{c}_0 = (0, 0)$, $\mathbf{c}_{1-4} = (\pm 1, 0), (0, \pm 1)$, and $\mathbf{c}_{5-8} = (\pm 1, \pm 1)$ (lattice units). The MRT collision step is

$$\mathbf{m}^* = \mathbf{m} - \mathbf{S}(\mathbf{m} - \mathbf{m}^{\text{eq}}), \tag{18}$$

where the moment vector $\mathbf{m} = \mathbf{M}\mathbf{f}$, $\mathbf{M}$ is the 9×9 transformation matrix from population to moment space, and $\mathbf{S} = \text{diag}(s_0, s_1, \ldots, s_8)$ with $s_\nu = s_\xi = 2/(1 + 6\nu_{\text{latt}})$ for the shear-stress relaxation rates, where $\nu_{\text{latt}} = U_{\text{ref}} L/(Re \cdot N)$ is the lattice viscosity (with N the grid size per cavity side and U_{ref} the reference lattice velocity). All other relaxation rates are

set to unity ($s_i = 1$ for non-viscous moments). The post-collision populations are streamed: $f_i(\mathbf{x} + \mathbf{c}_i, t + 1) = f_i^*(\mathbf{x}, t)$.

Lid boundary conditions are imposed using the Zou–He scheme: at the upper wall ($y = N$), the normal populations are updated to enforce the specified lid velocity $U_{\text{lid}}(x)$ while conserving local momentum. Bottom and lateral walls use standard half-way bounce-back boundary conditions.

8.2 S8.2 Ghia et al. Benchmark Validation

The MRT-LBM solver was validated against the reference benchmark data of Ghia et al. for the uncontrolled square cavity at $Re = 100$ and $Re = 1000$ using a 256×256 lattice. Main Sec. 4.2 reports the L_2 and L_∞ errors in the centerline velocity profiles. The errors are within the expected accuracy of the LBM at this resolution, confirming that the solver correctly reproduces the known benchmark solutions before being applied to the controlled-flow study.

8.3 S8.3 Control Transfer Protocol

The PINN-identified controls are transferred to the LBM solver by evaluating the control network output at the desired Re and discretizing the resulting $U_{\text{lid}}(x, t = 0, Re)$ profile onto the lattice lid nodes. For the stationary $A_2 \sin(2\pi x)$ and $\sin(\pi x)$ configurations, the lid velocity is constant in time and is applied identically at every LBM timestep. For the PINN mean control, the mean control profile is obtained by averaging the PINN control output $\{a_{mn}\}$ over the baseline training Re values and evaluating $U_{\text{lid}}(x, t = 0)$, yielding a stationary profile that is then applied as a steady lid condition in the LBM. No retraining or adaptation of the PINN is performed; the transfer is a one-time evaluation of the neural network.

The stationary approximation implicit in this protocol is quantified in Sec. S8.5. The B_1 branch carries a temporal energy fraction of about 7.4%, so the time-mean profile is not exactly the control that the PINN produces. When the full time-dependent Fourier control is transferred instead, its cycle-averaged dissipation at $Re = 500$ and $\bar{E}_r = 0.250$ is 1.2189×10^{-2}, about 2.5% above the 1.189×10^{-2} obtained from the time-mean profile, and its kinetic-energy fluctuation ratio is 0.04%. (The eight-phase quadrature of the temporal table gives 1.2370×10^{-2} at its native $\bar{E}_r = 0.2535$; brought to the same energy the two agree to 0.07%, so the small difference is an energy rescaling, not a quadrature error.) The temporal content of B_1 is therefore dynamically small but not negligible: it contributes a few per cent to the dissipation. The stationary transfer used in the main text is a controlled approximation, adequate for ranking the stationary controls, and Sec. S8.5 supplies the cycle-averaged value for the full control.

Norm versus peak velocity. The transfer fixes the quadratic actuation norm $E_{\text{act}} = \int_0^1 U_{\text{lid}}^2 \, dx$, not the peak wall velocity or the Reynolds number based on it. At $E^* = 0.25$ the rms wall velocity is 0.5 for every case, but the peak is $U_{\max} = 0.50$ for the sharp uniform lid, 0.52 for the tapered lid, 0.71 for the two harmonics and 0.78 for the B_1 mean profile, i.e. $Re_{\max} = U_{\max} Re =$ 250, 258, 354 and 390 at $Re = 500$. For the time-dependent branches the tabulated $U_{\max}$ is that of the time-mean profile, with the maximum over the period in parentheses; it is 0.04 (1.23) for B_2 and 0.62 (1.16) for B_3. The comparisons of the main text are therefore at equal quadratic norm and fixed viscosity, not at equal $Re_{\max}$, and the two conventions must not be mixed when the values are compared with lid-speed-based literature data.

Figure S4 draws the same four transfers at both Reynolds numbers. The ordering is identical at $Re = 500$ and $Re = 1000$: the uniform lid is the most dissipative, the stationary second harmonic the least, and the mean PINN profile and the first harmonic lie between them, with reductions of 76.5% and 82.3% for the second harmonic against 73.1% and 80.3% for the PINN mean profile. The near-equality of the last two bars is the observation that motivated the stationary reduction of the B_1 branch, and the velocity-scale caveat stated in Sec. S8.3 is the reason it is not reported as a performance claim in this study.

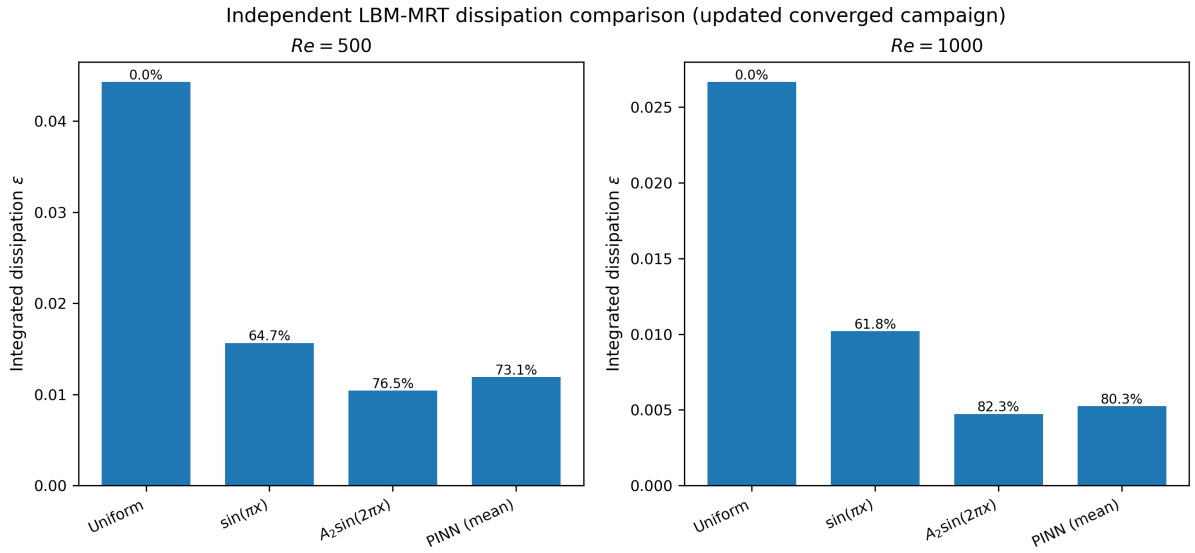


Figure S4: Integrated viscous dissipation of the four transferred configurations at $Re = 500$ (left) and $Re = 1000$ (right), with the percentage reduction relative to the uniform-lid bar written above each bar. These are the values of the dissipation table of Sec. 5.1 of the main text. The four profiles are compared at the same lid-velocity scale, not at the same realized actuation energy, whose ratio reaches four between the uniform lid and the second-harmonic control; the matched-energy ranking is the subject of Sec. S8.4.

8.4 S8.4 Grid convergence of the controlled-flow dissipation

The grid study of Main Sec. 4.3 was repeated for all four stationary controls (uniform lid, $k = 1$, $k = 2$ and the B_1 PINN mean profile) on the same three grids $N \in \{128, 256, 512\}$ at $Re = 500$, with the lid amplitude of each profile rescaled so that the actuation energy is the common value $E^* = \int_0^1 U_{\mathrm{lid}}^2\, dx = 0.25$. The uniform lid therefore uses $U_{\mathrm{lid}} = 0.5$, and the two harmonic controls use the amplitude $\sqrt{2E^*} = 0.7071$. All runs use the same MRT collision model, the same convergence criterion and up to 2×10^5 iterations. Because the lid amplitude differs from the $U_{\mathrm{lid}} = 1$ uniform baseline of the main text, the dissipation values reported here are *not* comparable with those of the problem statement; they are the values that define the iso-energetic comparison.

For each control the observed order of convergence is obtained from the Richardson extrapolation

$$p_{\mathrm{obs}} = \frac{\ln[(\varepsilon_{512} - \varepsilon_{256})/(\varepsilon_{256} - \varepsilon_{128})]}{\ln r}, \qquad r = 2, \tag{19}$$

the extrapolated value from

$$\varepsilon_\infty = \varepsilon_{512} + \frac{\varepsilon_{512} - \varepsilon_{256}}{r^{\,p_{\mathrm{obs}}} - 1}, \tag{20}$$

and the fine-grid grid-convergence index from

$$\mathrm{GCI}_{512} = F_s\, \frac{|(\varepsilon_{512} - \varepsilon_{256})/\varepsilon_{512}|}{r^{\,p_{\mathrm{obs}}} - 1}, \qquad F_s = 1.25. \tag{21}$$

When the three values are not monotone, when the estimated p_{obs} falls outside $[0.1, 4]$, or when the GCI exceeds 100%, the triple is flagged and its extrapolated value is not used: in that case the smooth-error assumption underlying the estimator is not credible.

The energy-matched values, the observed order of convergence, the extrapolated values and the fine-grid GCIs of the four stationary controls are collected in Table S12.

Table S12: Grid convergence and iso-energetic comparison at $Re = 500$ with matched actuation energy $E^* = 0.25$ ($E_r = \int_0^1 U_{\text{lid}}^2\, dx$). ε is the integrated viscous dissipation; GCI is the fine-grid Grid Convergence Index with $F_s = 1.25$. The B_1 PINN mean control is used as computed ($E_r = 0.2503$), i.e. without rescaling. A flagged GCI means that the triple does not satisfy the smooth-error assumption; this is the case for the uniform lid, discussed in the corner-singularity study.

Control	E_r	ε_{128}	ε_{256}	ε_{512}	GCI_{512} (%)	$\varepsilon_{512}/\varepsilon_{512}^{\text{unif}}$
Uniform lid	0.250	0.00774	0.00948	0.01102	130.71	1.000
$k = 1$ ($\sin \pi x$)	0.250	0.00635	0.00688	0.00713	3.85	0.647
$k = 2$ ($\sin 2\pi x$)	0.250	0.01012	0.01129	0.01185	5.62	1.076
B_1 PINN mean	0.250	0.01067	0.01189	0.01250	6.01	1.134

Table S13: Effect of the upper-corner velocity discontinuity on the grid convergence of the uniform lid at $Re = 500$, $E^* = 0.25$ and $N \in \{128, 256, 512\}$. The tapered lid has the same unit plateau but a raised-cosine taper $U = \frac{1}{2}[1 - \cos(\pi x/\delta)]$ of width $\delta = 0.05$ at each upper corner, so its tangential velocity is continuous and the amplitude is rescaled to the same $E_r = 0.25$. It is a different lid profile and does not replace the discontinuous lid in the iso-energetic ranking; it isolates the cause of the non-convergence.

Control	ε_{128}	ε_{256}	ε_{512}	p_{obs}	GCI_{512} (%)	$\varepsilon_{512}/\varepsilon_{512}^{\text{unif}}$
Uniform lid (discontinuous)	0.00774	0.00948	0.01102	0.18	130.71	1.000
Uniform lid, $\delta = 0.05$ taper	0.00717	0.00815	0.00866	0.96	7.79	0.786

The uniform lid is the weak point of this analysis, and the corner study reported in Table S13 identifies the cause. Its increments between successive grids are 1.745×10^{-3} and 1.540×10^{-3}, so they decay by only 12% per refinement, giving $p_{\text{obs}} = 0.18$ and $\text{GCI}_{512} = 131\%$. The formal Richardson limit is 2.25×10^{-2}, more than twice the finest-grid value; we quote it only to show that it is implausible, and do not use it. The cause is the tangential-velocity discontinuity at the two upper corners, which violates the smooth-error assumption and is absent from the sine controls. Replacing that lid by a cosine taper of width $\delta = 0.05$ at each corner, at the same energy, restores convergence: $p_{\text{obs}} = 0.96$, $\text{GCI}_{512} = 7.8\%$, $\varepsilon_\infty = 9.20 \times 10^{-3}$. The tapered lid is a different profile from the lid specified in the problem and is not substituted for it in the ranking; it serves to isolate the singularity and to provide a converged reference.

The corresponding convergence of the *ratios* $\varepsilon_{\text{control}}/\varepsilon_{\text{uniform}}$, which are the quantities used in Main Sec. 5.1, is reported separately because ratio errors need not cancel. The uniform lid has a discontinuous tangential velocity at the two upper corners, so its discretization error is dominated by a local singularity that is absent from the sine controls, whose lid velocity vanishes at both corners. The errors of numerator and denominator therefore have different origins and different signs of approach, and no error cancellation can be assumed a priori. Two normalizations are therefore reported in Table S14, one by the sharp lid specified in the problem and one by the corner-regularized lid.

With the sharp lid as reference the three ratio triples drift by 18–21% over the triple: $k = 1$ gives $0.821 \to 0.725 \to 0.647$ with $p_{\text{obs}} = 0.28$ and a formal GCI of 72%, while the $k = 2$ and B_1 triples are monotone but converge too slowly for the estimator ($p_{\text{obs}} = 0.038$ and 0.055, outside $[0.1, 4]$) and are flagged without a GCI. With the regularized lid as reference all three converge: $0.886 \to 0.843 \to 0.823$ ($p = 1.03$, GCI 3.1%) for $k = 1$, $1.413 \to 1.384 \to 1.368$ ($p = 0.82$, GCI 1.9%) for $k = 2$ and $1.488 \to 1.458 \to 1.443$ ($p = 0.97$, GCI 1.4%) for the B_1 mean profile, with extrapolated values 0.802, 1.347 and 1.427. The sign of the ordering is the same under both references and at every grid; the magnitude is not. A single-campaign GCI built on the sharp lid would therefore misrepresent the uncertainty of the iso-energetic comparison, and two

Table S14: Grid convergence of the dissipation ratio $\varepsilon/\varepsilon_{\mathrm{ref}}$ at $Re = 500$, $E^* = 0.25$, formed on each grid before the Richardson extrapolation, so the quoted GCI refers to the quantity used in the text. Two references are used: the sharp uniform lid specified in the problem, whose own dissipation grows by 16% between $N = 256$ and $N = 512$ and has no usable GCI, and a corner-regularized uniform lid (cosine taper $\delta = 0.05$, the corner table above) at the same energy. With the sharp reference all three ratios drift by 18–21% over the triple and two of them fall outside the admissible range of the estimator; with the regularized reference all three converge with $p_{\mathrm{obs}} \in [0.8, 1.1]$ and GCIs of 1.4–3.1%. A dash means the estimator is not applicable.

Reference	Control	ratio_{128}	ratio_{256}	ratio_{512}	p_{obs}	$\mathrm{GCI}_{\mathrm{ratio}}$ (%)	$\mathrm{ratio}_{\mathrm{exact}}$
sharp lid	$k = 1$ $(\sin \pi x)$	0.821	0.725	0.647	0.28	72.37	0.272
	$k = 2$ $(\sin 2\pi x)$	1.309	1.191	1.076	–	–	–
	B_1 PINN mean	1.379	1.254	1.134	–	–	–
regularized lid	$k = 1$ $(\sin \pi x)$	0.886	0.843	0.823	1.03	3.06	0.802
	$k = 2$ $(\sin 2\pi x)$	1.413	1.384	1.368	0.82	1.91	1.347
	B_1 PINN mean	1.488	1.458	1.443	0.97	1.39	1.427

independent checks are reported in its place.

The first is an inter-campaign cross-check. The same matched-energy problem had already been solved in an earlier, independent MRT-LBM campaign, driven with a different iteration budget: that campaign reduced the optimization residual to order 10^{-10} (4.8×10^5 iterations for the uniform case) where the present runs stop at the 2×10^5 cap with residuals of 3×10^{-6} to 3×10^{-5}. Its three stationary controls at $A_2 = 0.7071$, namely the uniform, $\sin(\pi x)$ and $\sin(2\pi x)$ lid profiles, are the uniform, $k = 1$ and B_1-mean cases of the present study at the same $E^* = 0.25$, so the two data sets are directly comparable: the six stationary values agree to 0.70% or better (uniform -0.18% at $N = 256$ and -0.70% at $N = 512$; $k = 1$ -0.18%, -0.64%; $k = 2$ $+0.03\%$, -0.38%), and the two transferred branch mean fields agree to 0.01% (B_2 seed 5: 3.0977×10^{-5} against 3.0979×10^{-5}; B_3 seed 0: 5.0510×10^{-3} against 5.0514×10^{-3}). The $N = 512$ ratios agree to 0.05% ($k = 1$) and 0.3% ($k = 2$). That the dissipation is insensitive to four orders of magnitude of optimization residual shows that the reported values are set by the discretization and by the physical configuration, not by the convergence history of the solver, and the inter-campaign spread is the reproducibility error quoted in Main Sec. 5.1.

The second check is the resolution trend itself. Under the regularized reference the two penalty ratios move towards unity as the grid is refined ($k = 2$: $1.413 \to 1.384 \to 1.368$; B_1 mean: $1.488 \to 1.458 \to 1.443$), so the iso-energetic penalty carried by the second harmonic decreases with resolution instead of being an artifact of one grid, whereas the benefit ratio of the first harmonic moves away from unity ($0.886 \to 0.843 \to 0.823$), i.e. its advantage grows. The remaining drift between $N = 256$ and $N = 512$ is 2.4%, 1.2% and 1.1% for the three ratios, consistent with the quoted GCIs. Neither the penalty nor the benefit is grid-independent, and the coarse-grid magnitudes must not be read as the asymptotic ones.

8.5 S8.5 Cycle-averaged response of the temporal controls

The two temporal controls, the full Fourier series and the modified-Chebyshev series, are periodic with the forced period $T = 2$. The PINN residual is imposed on the normalized interval $t \in [0, 1]$; the fundamental cosine factor $\cos(\pi t)$ therefore has period $T = 2$ in those units, and the control is exported on $t \in [0, 2]$, one full period, and advanced by 256 lattice steps. The dissipation of such a flow is not a single number: what a periodic actuation is charged for is the cycle average $\bar{\varepsilon} = T^{-1} \int_0^T \varepsilon(t)\, dt$, whereas the diagnostics of Sec. S8.4 report $\varepsilon(\langle u \rangle)$, the dissipation of the *time-mean* field. The two are not interchangeable. Because ε is a convex quadratic functional of the velocity gradient, Jensen's inequality gives $\bar{\varepsilon} \geq \varepsilon(\langle u \rangle)$, with equality only in the steady

limit; the gap is the dissipation carried by the fluctuations.

Table S15: Cycle-averaged response of the time-dependent controls at $N = 256$. $\bar{E}_r$ is the cycle-mean actuation energy, $\bar{\varepsilon}$ the cycle-averaged dissipation, A_ε the half-spread of $\varepsilon(t)$ over the cycle, and $R_K = K_{\mathrm{fluct}}/K_{\mathrm{mean}}$ the kinetic-energy fluctuation ratio. $\varepsilon(\langle u \rangle)$ is the dissipation of the time-mean field, shown for comparison. The dagger marks values obtained from the eight stored phases where the response is strongly unsteady; at $Re = 500$, where a direct accumulation is available, that quadrature is 5.4% low.

Re	Control	$\bar{E}_r$	$\bar{\varepsilon}$	A_ε	$\varepsilon(\langle u \rangle)$	R_K (%)
100	`pinn_t`	0.258	3.88×10^{-2}	2.8×10^{-3}	3.78×10^{-2}	0.15
500	`pinn_t`	0.250	1.22×10^{-2}	2.4×10^{-3}	1.17×10^{-2}	0.04
1000	`pinn_t`	0.269	1.13×10^{-2}	4.8×10^{-3}	5.19×10^{-3}	0.65
100	`cheb_t`	0.290	$1.51 \times 10^{-1\dagger}$	1.1×10^{-1}	2.42×10^{-4}	846.80
500	`cheb_t`	0.286	4.06×10^{-2}	3.5×10^{-2}	5.97×10^{-4}	43.80
1000	`cheb_t`	0.293	$2.01 \times 10^{-2\dagger}$	1.6×10^{-2}	5.56×10^{-5}	124.99

The cycle average is evaluated from the eight phase-averaged velocity fields stored by the solver, one every $T/8$. Because the response locks on the forcing frequency, the mean of eight equally spaced phases is an exact quadrature of the cycle average for every harmonic below the fourth, but the accuracy of the quadrature depends on how much energy the response puts in higher harmonics. This was checked directly, accumulating $\varepsilon(t)$ every fourth timestep over a window of one hundred forced periods after two hundred periods of transient. At $Re = 500$, for the quasi-steady Fourier control the direct average is 1.236×10^{-2} against 1.237×10^{-2} from the eight phases (0.07%), and the dissipation of the mean field agrees to 0.3%; the two estimates of the ratio $\bar{\varepsilon}/\varepsilon(\langle u \rangle)$ agree to 0.2% (1.042 against 1.040). For the strongly unsteady modified-Chebyshev control, whose fluctuations carry $R_K = 44\%$ of the kinetic energy, the direct average is 4.06×10^{-2} against 3.85×10^{-2} from the eight phases, i.e. the eight-phase value is 5.4% low: the higher harmonics that the eight phases alias are not negligible in this regime. The direct value is therefore used for $\bar{\varepsilon}$ of the Chebyshev control at $Re = 500$, and the eight-phase entries at $Re = 100$ and $Re = 1000$ are marked in Table S15 as lower estimates. The half-spread A_ε quoted in the table is the amplitude of $\varepsilon(t)$ over the cycle and is the uncertainty that a single cycle-mean number conceals.

Three points follow. First, for the full Fourier control the cycle average exceeds the mean-field value by only 3% at $Re = 100$ and 4% at $Re = 500$, so the 7.4% temporal content of the B_1 branch costs a few per cent of dissipation rather than a change of regime; at $Re = 1000$ the ratio rises to 2.19, i.e. the temporal contribution becomes comparable to the mean one. Second, the modified-Chebyshev controls are not low-dissipation states. Their mean-field values in Table S15 (2.4×10^{-4} to 6.0×10^{-4}) lie one to two orders of magnitude below every stationary control, but this is an artifact of time-averaging a strongly unsteady flow ($R_K = 44$–847%): the mean field retains almost no motion, while the cycle averages, 2.0×10^{-2} to 1.5×10^{-1}, are the largest of all the configurations examined. Third, the verdict on the two representations does not depend on which of the two functionals is reported: on the cycle average the Chebyshev controls are the most dissipative, not the least.

The iso-energetic ranking of Main Sec. 5.1 is therefore restricted to the stationary controls, for which a single time-independent dissipation value is well defined and for which the grid-convergence analysis of Sec. S8.4 applies. The temporal controls are reported here, separately, with their cycle mean and the amplitude of the oscillation around it.

A matched-energy ranking of the temporal branch representatives recovered by the optimization complements the stationary comparison of Sec. S8.4. The lid series of the independent-validation runs are normalized by the peak of their *time-mean* profile, a convention that fixes the amplitude of the stationary part only: the resulting cycle-mean actuation energies are 0.43

Table S16: Iso-energetic comparison of the four stationary controls and the temporal PINN branch realizations at $Re = 500$, all at $E^* = 0.25$ on the $N = 256$ lattice. For stationary controls $\varepsilon(\langle u \rangle) = \varepsilon$ and the cycle average coincides with it; for temporal controls both functionals are given (Jensen: $\bar{\varepsilon} \geq \varepsilon(\langle u \rangle)$). f_t is the temporal energy fraction of the lid control and A_ε the half-spread of $\varepsilon(t)$ over the cycle, i.e. the uncertainty concealed by a single cycle-mean number. The transfer uses one exported period of each realization, so the slow drift of the B_3 branch in the PINN (Section 3.3) is not represented here; the reported A_ε is the uncertainty of the cycle mean for that fixed realization.

Control	f_t	$\bar{E}_r$	$\varepsilon(\langle u \rangle)$	$\bar{\varepsilon}$	$\bar{\varepsilon}/\varepsilon(\langle u \rangle)$	A_ε
uniform lid	0.000	0.250	9.48×10^{-3}	9.48×10^{-3}	1.00	0.0×10^{0}
$\sin(\pi x)$	0.000	0.250	6.88×10^{-3}	6.88×10^{-3}	1.00	0.0×10^{0}
$A_2 \sin(2\pi x)$	0.000	0.250	1.13×10^{-2}	1.13×10^{-2}	1.00	0.0×10^{0}
B_1 mean field	0.000	0.250	1.19×10^{-2}	1.19×10^{-2}	1.00	0.0×10^{0}
B_1 full (Fourier series)	0.074	0.250	1.17×10^{-2}	1.22×10^{-2}	1.04	2.4×10^{-3}
B_2 seed 5 (max f_t)	0.998	0.250	3.10×10^{-5}	3.76×10^{-2}	1214.32	3.6×10^{-2}
B_2 seed 14 (median f_t)	0.975	0.250	1.82×10^{-4}	3.70×10^{-2}	203.52	3.5×10^{-2}
B_3 seed 0	0.448	0.250	5.05×10^{-3}	2.18×10^{-2}	4.32	1.5×10^{-2}
B_3 seed 4	0.637	0.250	3.12×10^{-3}	2.69×10^{-2}	8.62	2.6×10^{-2}

(B_1), 43.0 (B_2) and 0.42 (B_3), and the B_2 lid reaches 17 reference velocity units, outside the incompressible range of the solver. The transfers reported in Table S16 follow the convention of Main Sec. 2.2 instead: a single multiplicative factor brings the cycle-mean actuation energy to $E^* = 0.25$. Because the factor is uniform in space and time, f_2, f_t and C_3, and hence the branch label, are unchanged.

Four realizations of the $Re = 500$, $E^* = 0.25$ cell were transferred to the LBM with the per-step cycle-mean protocol above: B_2 seeds 5 and 14 ($f_t = 0.998$ and 0.975) and B_3 seeds 0 and 4 ($f_t = 0.448$ and 0.637).

The transfer is verified against the independent case study of Sec. S8.4: the mean-field dissipation of the B_2 seed-5 and B_3 seed-0 transfers reproduce the values of that earlier campaign to 0.01%, so the two independent solvers agree on the same lid series. On the cycle average, B_2 costs 3.76×10^{-2} and 3.70×10^{-2} (a within-branch spread of 1.7%), B_3 2.18×10^{-2} and 2.69×10^{-2} (a spread of 23%, i.e. the B_3 class is the less well defined of the two), and the full B_1 series 1.22×10^{-2}, i.e. the same order as its own mean field and as the stationary $k = 2$ and B_1-mean controls. The B_2 mean field, by contrast, dissipates 3.1×10^{-5} and 1.8×10^{-4}, four and two orders below its cycle average, and the cycle-mean kinetic energy of the seed-5 response is 2.47×10^{-4} for a mean field of 1.55×10^{-5}. The branch that the optimization accesses most readily at high temporal content is therefore the most dissipative one at matched energy, and the accessibility order of the branches is not a performance order.

Two caveats stay open. The transfers above keep the shape of a recovered control and rescale its amplitude, so they do not show what the optimizer would find under an explicit cycle-mean energy constraint in the loss; retraining under that constraint remains future work. And they are single-grid ($N = 256$) at $Re = 500$, so the three-grid study of Sec. S8.4 does not cover them. Each transfer is averaged over one hundred forced periods after two hundred transient periods and stopped when the drift between successive windows falls below 2×10^{-3}; the four branch transfers stopped at 1.28×10^{5} steps with drifts of 8.6×10^{-8} to 6.1×10^{-5}, while the quasi-steady Fourier control stopped at the largest drift of the set, 2.4×10^{-3}, which bounds the systematic error of its cycle mean. The representation comparison above uses the two PINN lid series, whose cycle-mean energies (0.25 and 0.29) are close to the target and require no rescaling.

8.6 S8.6 Field captures behind the main-text branch figures

The three field figures of the main text (Main Figs. 6–8) and Main Table 4 are built from a dedicated capture of one representative per branch, independent of the transfers of Table S16 and of the independent case study of Sec. S8.4. The three runs use the same solver, the same $N = 256$ lattice, the same $Re = 500$, the same period $T = 256$ steps and the same realised actuation energy. The lid series are the exported PINN Fourier coefficients with a single multiplicative factor, 0.9930 for B_1 ($E_r = 0.2500$), 1.0000 for B_2 seed 5 ($E_r = 0.2504$) and 1.0000 for B_3 seed 0 ($E_r = 0.2503$). Only the B_1 factor differs from unity, by 0.7%, and it rescales the effective Reynolds number of that run by the same 0.7%; no other run is rescaled, and the realised rather than the prescribed actuation energy is what the figures compare.

Each run was advanced for two hundred forced periods before a drift test on the sampled dissipation $\varepsilon(t)$, which compared two sliding windows of one hundred periods sampled every fourth step, and then for sixty-four forced periods of sampling. All three settled reaches ended at the five-hundred-period cap, with relative drifts over the last window of 1.5×10^{-3} (B_1), 2.6×10^{-9} (B_2) and 7.3×10^{-5} (B_3), after 1.28×10^5 iterations each. The longer settling, compared with the two-hundred-period protocol of the transfers above, is what resolves the B_3 drift at the 10^{-4} level. Sampling covers sixty-four periods, i.e. eight blocks of eight cycles, which are the blocks behind the uncertainties of Main Table 4.

The cycle means reproduce the earlier independent campaign: the dissipation is 1.2194×10^{-2}, 3.7616×10^{-2} and 2.1826×10^{-2} against 1.2189×10^{-2}, 3.7616×10^{-2} and 2.1824×10^{-2}, the largest difference being 0.04%, and the mean fields agree to 10^{-5}. The cycle-mean kinetic energy of the B_3 capture is 3.98×10^{-3}, 1.3% above the 3.93×10^{-3} of the earlier eight-phase quadrature; that quadrature is the less accurate estimate, since it is 5.4% low for B_2, so the new value is the one quoted here and in the main text.

Two caveats apply to every figure built on these captures. First, they are single realizations. B_2 seed 5 lies 0.021 from the centroid of its class in (f_2, f_t), against a median distance of 0.014 and a class radius of at most 0.028, and it dissipates 1.7% more than seed 14. B_3 seed 0 sits at the low-dissipation end of a class that has only two members in this cell, seed 4 dissipating 23% more. The figures therefore show a typical member of B_2 and one pole of a loosely defined B_3, never a class average, and the captions label them as single realizations. Second, the fields are captured at one grid ($N = 256$) and one Reynolds number ($Re = 500$); the three-grid study of Sec. S8.4 covers the stationary controls, not these temporal ones. The drift, the periodicity check below and the block uncertainties bound the numerical error of a cycle mean at fixed realization; they say nothing about the spread of a class, which is what the iso-energetic table reports.

8.7 S8.7 Wall-shear estimator and closure uncertainty

The identity $\overline{P}_{\mathrm{lid}} = \overline{D}$ of the main text is exact for the continuous equations, so any residual measured between the two cycle means is a property of the discrete estimators, not of the physics. Three choices enter the estimate of P_{lid}: the velocity used at the wall node, the order of the one-sided wall-normal derivative, and the number of phases used to average the product $U_{\mathrm{lid}}\tau_w$.

Wall velocity. The transfer imposes U_{lid} through the Zou–He condition, and the velocity actually recovered at the wall node differs from the prescribed value by the discrete slip of the scheme. Using the recovered velocity $u_x(N-1, \cdot)$ in the product, rather than the prescribed U_{lid}, removes most of the residual. With a first-order one-sided derivative the cycle-mean residual $\overline{P}_{\mathrm{lid}}/\overline{D}-1$ is −2.89%, −0.08% and −0.87% for B_1, B_2 and B_3 when the recovered velocity is used, against +5.30%, +55.40% and +43.56% when the prescribed profile is used. The prescribed-velocity estimate is therefore discarded: it measures the slip of the boundary condition, not the power transmitted by the wall. All main-text values use the recovered velocity.

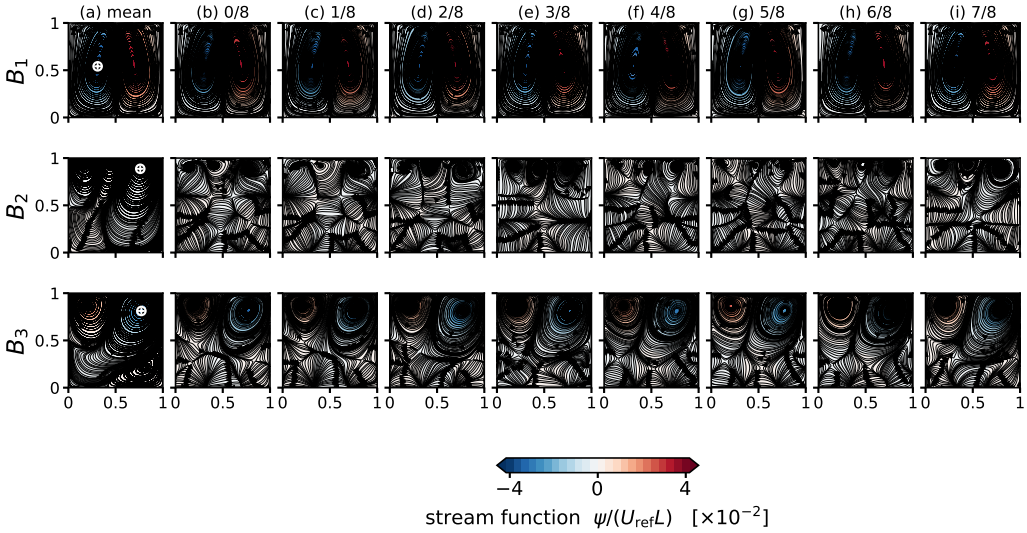


Figure S5: All eight phase bins of the three captures of Main Figs. 6–8, on the same common stream-function scale and contour levels as the main figure. Column (a) is the time mean, columns (b)–(i) the phase averages at $t/T = 0,\ 1/8, \ldots, 7/8$. The phase averages of the first and second halves of the sampling window differ by at most 1.0×10^{-3} in relative field norm for the three branches, so the state is periodic and the four phases selected for the main figure are representative rather than chosen for appearance. B_1 is stationary to 2.7% throughout the cycle; B_2 shows that its instantaneous structure is not a small perturbation of its mean, the instantaneous stream-function extrema being one to three times the mean one, while its mean circulation is 19 times weaker. Streamlines are tangent to the instantaneous velocity and are not particle paths.

Derivative order. Replacing the first-order one-sided derivative by the three-point second-order form $(3u_{N-1} - 4u_{N-2} + u_{N-3})/(2\Delta x)$ moves the mean-field residual of B_1 from -2.87% to $+0.58\%$, i.e. it crosses zero, which shows that the truncation error of the first-order estimate is of the order of the residual itself. For the cycle means the same change gives $+1.36\%$ for B_1, but $+21.5\%$ for B_2 and $+16.6\%$ for B_3: the three-point stencil amplifies lattice-scale noise in the thin Stokes layer of the strongly oscillating lids, where successive wall-normal velocities differ by more than the signal. The second-order form is therefore better for the smooth quasi-stationary case and worse for the unsteady ones, and no single choice is optimal for all three branches. The first-order form is retained for consistency across the table, and the residual is quoted as measured.

Phase quadrature. The eight stored phase fields are sufficient for the dissipation, which is a quadratic functional of the instantaneous field, but not for the power, which is a product of two quantities that both vary within a phase bin. Averaging the recovered-velocity, first-order power over the eight phases gives -0.27%, -5.39% and -4.46% relative to the direct accumulation over 4096 steps for B_1, B_2 and B_3. The direct accumulation is therefore used for $\overline{P}_{\mathrm{lid}}$ in the main text, and the eight-phase value is not substituted for it.

Taken together, the three checks bound the estimator uncertainty at a few per cent for the quasi-stationary branch and at roughly ten per cent for the strongly unsteady ones. This is why the main text presents $\overline{P}_{\mathrm{lid}} = \overline{D}$ as an energetic verification with an explicit estimator uncertainty, and why the conclusions rest on differences of a factor of three rather than on the third significant digit of the budget.

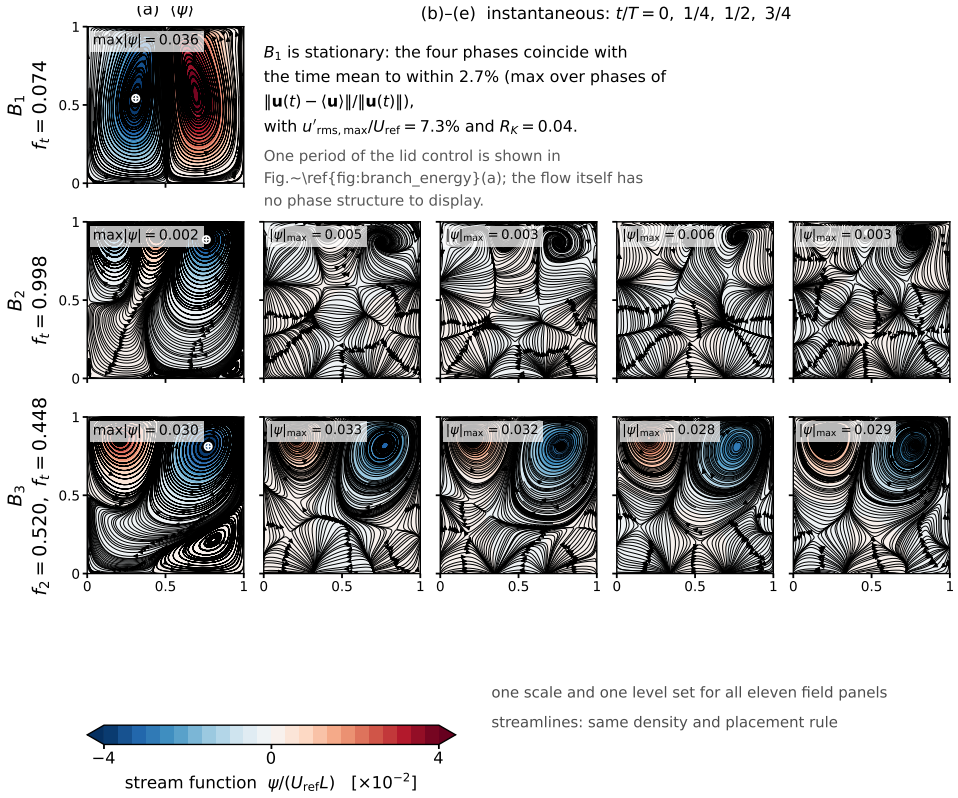


Figure S6: The layout of Main Fig. 6 with a separate stream-function scale in every panel, annotated with $\max|\psi|$ (mean panels; the instantaneous panels are annotated with their own $|\psi|_{\max}$). The common scale of the main figure is what makes the B_2 mean cell nearly invisible; this version shows its shape—three extrema, all within the top tenth of the cavity, with a reversed pair under the lid—and shows that the B_3 mean circulation is concentrated under the lid rather than in the mid-depth cells of B_1. The two versions are complementary and must be read together: only the common scale supports a quantitative comparison between branches.

9 S9. Symbolic Regression Configuration and LOCO Details

9.1 S9.1 PySR Configuration

Each fit—the six leave-one-condition-out refits of Sec. S9.3—used the same configuration, listed in Table S17.

Table S17: PySR hyperparameter configuration, identical for every refit.

Parameter	Value
Populations	20
Iterations	100 per population
Binary operators	$\{+, -, \times, /\}$
Unary operators	$\{\sin, \cos, \exp, \text{square}, \text{sqrt}\}$
Loss	$L2$ distance
Parsimony coefficient	0.01
Maximum expression size (nodes)	30
Model selection	Best accuracy on the Pareto frontier
Random seed	42, deterministic, serial parallelism
Backend	Julia with SymbolicRegression.jl

The search is run on standardized inputs and targets, and each returned expression is converted back to the physical variables x and Re before it is evaluated, so that the constants quoted in Sec. S9.3 apply directly to (x, Re) without rescaling.

9.2 S9.2 Input-Output Specification

The input dataset consists of 1 200 (x, Re) sample pairs, i.e. a 200-point uniform grid on $x \in [0, 1]$ at each of $Re \in \{100, 300, 500, 700, 900, 1000\}$, restricted to B_1-classified realizations from the broad campaign. The intermediate Reynolds values $Re \in \{400, 600, 800\}$ are not populated by any B_1 realization in the campaign and are therefore excluded from the cross-validation. The target output is the branch-representative stationary control profile, taken as the time average of $U_{\text{lid}}(x, t; Re)$ over the sampling window $t \in [0, 2]$ on 64 time samples, extracted at a fixed actuation energy $E^* = 2.0$; among the sampled energy slices, $E^* = 2.0$ is the only one that populates all six Reynolds conditions with B_1 realizations. The time-averaged profile is used because the B_1 branch is dominated by the $n = 0$ (stationary) mode, so the time mean fully characterizes the leading spatial structure while suppressing residual temporal content. At each condition the representative (largest f_2) realization is used; the representatives are the seeds listed in Table S18.

The regressed quantity is the *peak-normalized* profile, $\bar{U}_{\text{lid}}(x; Re)/\max_x |\bar{U}_{\text{lid}}(x; Re)|$, so the expression returned by a fit is a shape and carries no amplitude. The dimensional profile is recovered by multiplying by the peak scale of the corresponding condition, listed in Table S18. Because the target is a time mean, no time variable appears in any of the returned expressions.

9.3 S9.3 LOCO Validation Results

PySR is a stochastic symbolic search: each refit returns the best expression of its own run, and the six refits are therefore six different expressions rather than six evaluations of one formula. Tables S18 and S19 report, for each fold, the metrics of the expression that the fit returned and that expression exactly as recorded.

The LOCO R^2 ranges from 0.8238 at the lowest Reynolds number ($Re = 100$) to 0.9718 at $Re = 500$, with a mean of 0.914. The $Re = 100$ fold is the most demanding extrapolation case: it lies at the low-Re edge of the campaign, so that only the higher side ($Re = 300$) is available for interpolation. Three of the six fits ($Re = 300, 700, 1000$) return the same nested argument

Table S18: Per-fold metrics of the recorded $E^* = 2.0$ LOCO fits. R^2 and the root-mean-square error (RMSE) refer to the held-out peak-normalized profile on its 200-point grid. "Peak scale" is $\max_x |\bar{U}_{\mathrm{lid}}|$ of the data profile, which converts an expression into the dimensional mean profile. $E_r(\cdot) = \int_0^1 (\cdot)^2\, dx$ of that dimensional profile, evaluated for the data and for the fold expression, and their relative difference, are the amplitude error left by the description.

Held-out Re	Seed	f_2	R^2	RMSE	Peak scale	E_r^{data}	E_r^{SR}	ΔE_r
100	3	0.884	0.8238	0.252	2.269	1.872	1.562	−16.6%
300	2	0.738	0.8984	0.196	2.174	1.811	2.001	+10.5%
500	2	0.866	0.9718	0.113	2.045	1.918	1.686	−12.1%
700	13	0.943	0.9125	0.202	2.002	1.880	1.661	−11.7%
900	1	0.739	0.9283	0.177	1.923	1.620	1.452	−10.4%
1000	1	0.936	0.9518	0.143	2.148	1.967	2.093	+6.4%

$3.4468\,x - 1.7234$, whose phase shift is exactly half a wavelength and which is therefore antisymmetric about the cavity centre; this recurrence, not a shared closed form, is the reproducible structural content of the symbolic analysis. The amplitude is not reproduced: the energy of the dimensional profile rebuilt from a fold expression differs from that of the data profile by −16.6% to +10.5%, consistent with the fact that the regression target is a normalized shape. The result therefore shows that the B_1 control family is described, fold by fold, by compact expressions across the six conditions at which it is reliably recovered, while the reduced fidelity at the low-Re edge is consistent with the weakening B_1/B_2 separation reported in the main text. The coherence is, moreover, energy-dependent: repeating the LOCO on the $E^* = 0.5$ and $E^* = 1.0$ slices yields high scores at most folds but extrapolation failures at the corresponding edge conditions ($Re = 100$ for $E^* = 0.5$, and $Re = 700$ and $Re = 1000$ for $E^* = 1.0$). The $E^* = 2.0$ slice is thus the only one yielding a uniformly positive LOCO, and is the one reported in Tables S18 and S19 and in Sec. S9.3.

10 S10. Reproducibility and Computational Pipeline

10.1 S10.1 Software Environment

Table S20 lists the software environment used for all the computations reported here and in the main text.

10.2 S10.2 Hardware

PINN training was performed on a single NVIDIA A100 GPU (80 GB HBM2e) with double-precision computation enabled. The custom C++/OpenMP MRT-LBM solver was executed on a multi-core CPU workstation (8 cores, 32 GB RAM) with OpenMP thread parallelization over lattice rows.

10.3 S10.3 Computational Cost

Typical wall-clock times for the main computational components are:

- One PINN realization (300 pre-training + 3 000 joint epochs): approximately 45 minutes on a single A100 GPU.
- LBM simulation to steady state at $N = 512$: approximately 2 hours on 8 CPU cores.
- Full 185-realization broad campaign: approximately $185 \times 45 \approx 139$ GPU-hours (parallelized across 4 GPUs: approximately 35 hours wall time).

Table S19: The six fold expressions of the $E^* = 2.0$ LOCO, recorded exactly as returned after conversion to (x, Re). In this monospace notation `*` is multiplication and `**` is exponentiation; `x` and `Re` are the only variables. The expression held out at $Re = 500$, the best-scoring fold, contains no Re at all, which is the clearest indication that these are per-condition descriptions rather than one control law.

Held-out Re	Expression
100	`-0.656531744300912*sin(0.00044303087755444 3*Re+8.35334751283491*x-4.7861 1502315448)-0.656531744300912*sin(0.00029414433956932 1*Re+2.601491007477 48*x-(0.00390434404721515*Re-2.6549539521063)**2*(0.259675841445626*x-0. 129837920722813)-1.50076365464588)+0.0835326911204075`
300	`0.0355674592281737-0.94948430325695*sin(4.33312505481874*x+sin(3.4468241 9263376*x+0.24729648*sin(5.87977914011404*exp(sin(0.003125*Re-2.0)))-1.9 1762330631688)-2.25663605340937)`
500	`0.038002190814428-0.927540972495295*sin(3.4455988*sin(2.27531862008866*x -1.20059356004433))`
700	`(-0.640381176720972*(1.62396632386904*x-0.948239180502976)*(3.4468241926 3376*x-1.72341209631688)+1.1369819864629)*sin(-6.24748466244156*x+0.4038 995*cos(-0.00290619096859548*Re+1.4645397440588*x+0.895197070384071)+3.1 2374233122078)+0.0421374749638329`
900	`-1.40941323554341*x+0.645040218519208*sin(-8.63782139306147*x+0.4854202* sin(0.0032009219983224*Re-1.23114080131396*x+0.886964461529331)**2+4.318 91069653074)+0.749507753884623`
1000	`-1.39188046572375*x+0.646718514892542*sin((0.00039947948105659 3*Re-2.710 1076405283)*(3.44682419263376*x-1.72341209631688)+0.31353217)+0.78394993 9083173`

Table S20: Software environment for all numerical computations.

Component	Version
Python	3.10.12
PyTorch	2.0.1+cu118
NumPy	1.24.3
SciPy	1.11.1
PySR	0.16.3
Julia	1.9.2
GCC (for LBM solver)	11.4.0
OpenMP	4.5

- λ_{var} campaign (90 realizations): approximately 68 GPU-hours.

10.4 S10.4 Random Seed Control

Each realization is assigned a unique integer seed $s \in \{0, 1, \ldots, N_s - 1\}$. At the beginning of each realization the seed is passed to the pseudorandom generators of the deep-learning framework, of the array library and of the standard random module, and, for multi-GPU runs, to the device generators as well. Each realization is therefore strictly reproducible from its seed number.

10.5 S10.5 Inventory of Numerical Experiments

Table S21 lists every numerical campaign reported in the main text and in this document, with the method used and the number of runs. The purpose is to make the campaign structure auditable: the run counts are stated per study and are not summed, because the branch-capture representatives reappear among the iso-energetic transfers, and because the capacity, truncation and aspect-ratio studies re-run the reference protocol with a single factor changed.

Table S21: Inventory of the numerical experiments reported in this article. Runs are counted per campaign. Several studies re-use the same trained control, the same seeds or the same lattice fields, so the column is not additive and no deduplicated grand total is claimed.

Campaign	Method	Runs	Reported in
Broad (Re, E^*) campaign	PINN	185	Main Sec. 3.2; Table S3
Variance-regularization sweep	PINN	90	Sec. S4; Tables S4 and S7
Ten-seed multiplicity	PINN	10	Table S5
Network capacity	PINN	3	Table S9
Control truncation	PINN	6	Table S10
Cavity aspect ratio	PINN	3	Table S11
Fourier vs. modified Chebyshev	PINN	6	Sec. S7.2
Stationary iso-energetic	LBM-MRT	12	Main Sec. 5.1; Table S12
Corner-taper control	LBM-MRT	3	Table S13
Temporal cycle mean	LBM-MRT	6	Table S15
Branch capture	LBM-MRT	3	Main Secs. 6.3–6.4; Main Table 4
Branch iso-energetic transfers	LBM-MRT	4	Table S16
LOCO symbolic refits	PySR	6	Tables S18 and S19
Classification sensitivity	reanalysis	0	Table S2
Dissipation-ratio GCI	derived	0	Table S14

The two zero-run rows are re-analyses of runs already listed: the classification sensitivity relabels the 185 broad-campaign outcomes, and the ratio GCI is formed from the dissipation values of the iso-energetic campaign. The stationary iso-energetic row counts four controls on three grids; the corner-taper row adds a fifth profile on the same three grids. The LOCO row counts symbolic fits, not simulations.

10.6 S10.6 Data Availability

The cell-aggregated record of all 185 runs—realization counts, branch frequencies, modal statistics and energy calibration per cell—is reported in Supplementary Table S3; the seed-level records retained for the multiplicity and ablation studies are reported in Tables S5 and S6. Further details are available from the authors upon reasonable request.